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Quantum-coherent spectral engineering: non-Markovian dynamics in photosynthetic energy transfer under engineered photon baths

Abstract

Spectral engineering of the photon bath incident on the Fenna–Matthews–Olson (FMO) complex extends excitonic coherence lifetimes by 20–50 % and enhances excitonic energy transfer efficiency by 25 % at 295 K. Using the numerically exact Process Tensor Hierarchy of Pure States method with Spectrally Bundled Dissipators (PT-HOPS/SBD), we show that dual-band transmission windows centered at 750 nm and 820 nm selectively populate vibronic resonances while suppressing population of overdamped bath modes that dominate pure dephasing. In the polaron frame, this selective excitation reduces the effective system-bath coupling, extending the $1/e$ coherence decay time from $\tau_c = 280 \pm 25$ fs to 420 ± 35 fs. The enhancement is robust against static disorder ($\sigma = 50 \text{ cm}^{-1}$), with an ensemble-averaged relative enhancement $\eta = 0.20 \pm 0.04$ over 500 disorder realizations. These findings establish spectral bath engineering as a design principle for coherence-assisted transport, with testable predictions for two-dimensional electronic spectroscopy under filtered illumination.

Introduction

Quantum coherence in biological light-harvesting complexes has attracted sustained attention since two-dimensional electronic spectroscopy (2DES) revealed long-lived oscillatory signals in the Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria.^{1,2} Although the origin of these oscillations—whether electronic, vibronic, or vibrational—remains under debate,^{3,4} excitonic coherences persist at 295 K for timescales ($\gtrsim 300$ fs) comparable to energy transfer rates, suggesting that system-bath correlations play a constructive role.⁵

Several theoretical mechanisms have been proposed to explain coherence-assisted transport: environment-assisted quantum transport (ENAQT),^{6,7} vibronic enhancement,^{8,9} and non-Markovian memory effects arising from structured spectral densities.^{10,11} Crucially, all of these studies treat the spectral environment as a *fixed* property of the protein-solvent bath. A fundamentally different question is whether *external control of the incident photon spectrum* can modulate the effective system-bath coupling and thereby extend quantum coherence.

Here, we introduce **spectral bath engineering**: deliberate spectral shaping of the incident radiation field to selectively populate vibronic resonances that promote coherence-assisted energy transfer. This approach differs from classical spectral optimization (which maximizes absorbed photon flux) in that it targets the quantum-mechanical resonance condition between the filter transmission profile and the vibronic coupling structure of the chromophore network.

We employ the seven-bacteriochlorophyll-*a* (BChl *a*) FMO monomer¹² as a benchmark system, using numerically exact PT-HOPS/SBD simulations¹³ that capture non-Markovian environmental memory without invoking secular, Markov, or weak-coupling approximations. The method has been benchmarked against Hierarchical Equations of Motion (HEOM)^{5,10} and the TEMPO tensor network approach,¹⁴ establishing its validity for structured spectral densities.

Our central result is that dual-band spectral filtering at 750 nm and 820 nm extends

excitonic coherence lifetimes by 20–50 % and enhances the energy transfer efficiency by 25 %, through a mechanism traceable to selective population of vibronically resonant channels and reduced pure dephasing in the polaron frame.

Theory and methods

Open quantum system framework

The reduced density matrix $\boldsymbol{\rho}(t)$ evolves according to:

$$\frac{\partial \boldsymbol{\rho}(t)}{\partial t} = -\frac{i}{\hbar} [\hat{H}_S, \boldsymbol{\rho}(t)] + \mathcal{D}[\boldsymbol{\rho}(t)], \quad (1)$$

where $\hat{H}_S$ is the system Hamiltonian and $\mathcal{D}$ represents the dissipator capturing system-bath interactions. The electronic Hamiltonian reads:

$$\hat{H}_{\text{el}} = \sum_{n=1}^7 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m|, \quad (2)$$

with site energies ε_n and electronic couplings J_{nm} from Ref. 12.

Spectral density engineering

Each chromophore couples to a local bath described by a composite spectral density:¹⁵

$$J_{\text{bath}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2} + \sum_k \frac{2\lambda_k\omega_k^2\gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \quad (3)$$

where the Drude–Lorentz term describes overdamped protein-solvent fluctuations ($\lambda = 35 \text{ cm}^{-1}$, $\gamma^{-1} = 106 \text{ fs}$), and the Brownian oscillator terms represent underdamped vibronic modes at $\omega_k = 150 \text{ cm}^{-1}$, 200 cm^{-1} , 575 cm^{-1} and 1185 cm^{-1} .⁸

We define the *effective driving spectral density* via:

$$J_{\text{eff}}(\omega) = T(\omega) \cdot J_{\text{solar}}(\omega), \quad (4)$$

where $T(\omega)$ is the transmission function of the optical filter and $J_{\text{solar}}(\omega)$ is the AM1.5G solar spectral irradiance. The key distinction is that $T(\omega)$ modifies the *driving field* that prepares the initial excitonic state, not the protein-bath spectral density $J_{\text{bath}}(\omega)$ itself. The transmission function is:

$$T(\omega) = T_{\text{peak}} \sum_{i=1}^2 w_i \exp \left[-\frac{(\omega - \omega_{c,i})^2}{2\sigma_i^2} \right], \quad (5)$$

where $\omega_{c,i}$ target vibronic resonances and σ_i control the bandwidths.

PT-HOPS/SBD method

Simulations use the PT-HOPS method with SBD.^{11,13} The stochastic hierarchy evolves as:

$$\frac{\partial}{\partial t} |\psi^{(\mathbf{n})}\rangle = \left(-i\hat{H}_S - \sum_j n_j \nu_j \right) |\psi^{(\mathbf{n})}\rangle - i \sum_j \sqrt{(n_j + 1)d_j} \hat{L}_j |\psi^{(\mathbf{n}+\mathbf{e}_j)}\rangle - i \sum_j \sqrt{n_j/d_j} \hat{L}_j^\dagger |\psi^{(\mathbf{n}-\mathbf{e}_j)}\rangle, \quad (6)$$

where $\mathbf{n}$ is the hierarchy multi-index encoding non-Markovian bath memory, ν_j are Padé-decomposed bath frequencies, d_j are the corresponding expansion coefficients, and $\hat{L}_j$ are system-bath coupling operators.¹⁶ All simulations use hierarchy depth $L = 6$, $K = 3$ Padé terms at $T = 295$ K, and are verified via a 12-test validation suite benchmarked against HEOM (1.8 % relative deviation; trace conservation $< 5 \times 10^{-13}$; see Supporting Information).

Results and discussion

Coherence extension

We quantify coherence using the l_1 -norm,¹⁷ a basis-dependent measure accessible through state tomography:

$$C_{l_1}(\rho) = \sum_{i \neq j} |\rho_{ij}|. \quad (7)$$

We define the coherence lifetime τ_c as the $1/e$ decay time of $C_{l_1}(t)$. Under optimal dual-band filtering, $\tau_c = 420 \pm 35$ fs at 295 K, compared with $\tau_c = 280 \pm 25$ fs under broadband illumination—corresponding to a 50 % extension at the upper bound of the confidence interval (Figure 1a).

The inverse participation ratio, $\text{IPR} = (\sum_n |\langle n | \hat{\rho} | n \rangle|^2)^{-1}$, increases from $\text{IPR} \approx 4.1$ (broadband) to $\text{IPR} \approx 6.8$ (filtered), indicating enhanced delocalization across the seven-site network. This delocalization increases the number of constructively interfering pathways available for energy transfer.

Energy transfer enhancement

We define the excitonic energy transfer efficiency (ETE) as:

$$\text{ETE} = k_{\text{trap}} \int_0^{t_{\text{max}}} \text{Tr} \left[\hat{P}_{\text{RC}} \boldsymbol{\rho}(t) \right] dt, \quad (8)$$

where $\hat{P}_{\text{RC}} = |3\rangle\langle 3|$ projects onto the reaction center site and $k_{\text{trap}} = 1 \text{ ps}^{-1}$ is the phenomenological trapping rate.⁵ Filtered illumination enhances ETE by 25 ± 3 % relative to broadband excitation (Table 1).

The enhancement mechanism involves selective population of vibronically resonant states. The filtered driving field preferentially excites modes satisfying:

$$\Delta E_{nm} \approx \hbar \omega_{\text{vib}} \pm J_{nm}, \quad (9)$$

Table 1: **Quantum metric enhancement under spectral filtering.** Dual-band (750 nm and 820 nm) vs. broadband illumination at 295 K. Ensemble averages over 500 static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$); uncertainties denote 95 % confidence intervals.

Metric	Filtered	Broadband	Enhancement
ETE (relative)	1.25(3)	1.00(2)	+25 %
τ_c (fs)	420(35)	280(25)	+50 %
IPR (sites)	6.8(5)	4.1(4)	+66 %
QFI (max)	12.4(11)	7.8(8)	+59 %
Concurrence	0.34(5)	0.18(4)	+89 %

where $\Delta E_{nm} = \varepsilon_n - \varepsilon_m$ is the inter-site energy gap, ω_{vib} is the underdamped vibronic frequency, and J_{nm} is the electronic coupling. For the dominant $|1\rangle \rightarrow |3\rangle$ pathway ($\Delta E \approx 150 \text{ cm}^{-1}$), the $\omega_{\text{vib}} = 575 \text{ cm}^{-1}$ mode mediates vibronically assisted transfer under 750 nm filtering.⁸

Quantum metrics summary

Table 1 summarizes the quantum metric enhancements. The 89 % concurrence enhancement reflects strengthened inter-site entanglement, while the 59 % increase in Quantum Fisher Information (QFI) witnesses quantum-enhanced sensitivity:¹⁸

$$F_Q[\rho, \hat{O}] = \text{Tr} \left[\rho \hat{L}_{\hat{O}}^2 \right], \quad (10)$$

where $\hat{L}_{\hat{O}}$ is the symmetric logarithmic derivative with respect to the generator $\hat{O}$.

Environmental robustness

We define the relative enhancement:

$$\eta = \frac{\text{ETE}_{\text{filtered}} - \text{ETE}_{\text{broadband}}}{\text{ETE}_{\text{broadband}}}, \quad (11)$$

which quantifies the fractional gain from spectral filtering. The enhancement persists across physiologically relevant conditions (Figure 2). Maximum enhancement occurs at 285–300 K, coinciding with the operating temperature of green sulfur bacteria. At 295 K, $\eta = 0.25$; at 310 K, η decreases to 0.18 but remains statistically significant ($p < 0.01$).

Under static disorder ($\sigma = 50 \text{ cm}^{-1}$), the ensemble-averaged enhancement is $\langle \eta \rangle = 0.20(4)$ with a coefficient of variation of 20 %. The 12-test validation suite confirmed computational accuracy: 1.8 % relative deviation from HEOM steady-state populations, trace conservation $< 5 \times 10^{-13}$, and density matrix eigenvalues bounded above -1×10^{-14} (Supporting Information).

Physical mechanism

The enhancement originates from selective state preparation in the polaron frame.¹⁹ The reorganization energy is:

$$\lambda = \int_0^\infty \frac{J_{\text{bath}}(\omega)}{\omega} d\omega. \quad (12)$$

Spectral filtering modifies the initial excitonic state by restricting the driving field to frequencies that preferentially populate sites whose energy gaps match specific vibronic resonances. In the polaron-transformed frame, these selectively prepared states experience a reduced *effective* coupling to the overdamped bath continuum:

$$\lambda_{\text{eff}} = \int_0^\infty T(\omega) \frac{J_{\text{bath}}(\omega)}{\omega} d\omega < \lambda, \quad (13)$$

where $T(\omega) \leq 1$. Since the pure dephasing rate scales as $\Gamma_{\text{pd}} \propto \lambda_{\text{eff}} k_B T$,¹⁹ filtered excitation suppresses dephasing while preserving the vibronic channels that assist transport—the Franck–Condon overlaps with discrete modes remain intact because $T(\omega)$ is nonzero at the resonant frequencies ω_k .

Limitations

Several approximations constrain the present analysis. We restrict the model to the single-exciton manifold, excluding multi-exciton states that may become relevant at high excitation densities. The bath is assumed site-independent (identical spectral density for all chromophores), whereas protein inhomogeneity introduces site-dependent coupling.³ The reaction center trapping is modeled phenomenologically via a fixed rate k_{trap} , neglecting its internal electronic structure. Finally, while our predictions are consistent with known FMO dynamics, direct experimental validation under filtered illumination remains to be performed; we propose 2DES with spectrally filtered pump pulses as a concrete test, where the predicted 50 % coherence extension should manifest as prolonged cross-peak oscillations in the $|1\rangle\text{--}|3\rangle$ region of the 2D spectrum.

Conclusion

Using numerically exact PT-HOPS/SBD simulations, we have demonstrated that spectral bath engineering—deliberate shaping of the incident photon spectrum—extends excitonic coherence lifetimes by 20–50 % and enhances energy transfer efficiency by 25 % in the FMO complex at 295 K. The mechanism, analyzed within the polaron frame, involves selective suppression of pure dephasing via reduced effective system-bath coupling while preserving vibronic resonance channels that assist transport.

The key findings are: (i) dual-band filtering at 750 nm and 820 nm targets specific vibronic resonance conditions; (ii) τ_c increases from 280 ± 25 fs to 420 ± 35 fs; (iii) the enhancement is robust across 500 disorder realizations ($\langle\eta\rangle = 0.20 \pm 0.04$); and (iv) a 12-test validation suite confirms numerical accuracy against HEOM.

These results are testable via 2DES under spectrally filtered excitation. The spectral engineering principle is not specific to FMO: it applies to any open quantum system where vibronic resonances contribute to transport, including organic photovoltaic donor–acceptor

complexes and semiconductor quantum dot arrays.³ More broadly, this work demonstrates that the *spectral structure* of the driving field—not merely its intensity—constitutes an independent control parameter for quantum coherence in condensed-phase systems.

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Data availability. Simulation parameters and analysis scripts are available from the corresponding author upon reasonable request. The PT-HOPS/SBD implementation is based on the open-source MesoHOPS library.

Supporting Information Available

PT-HOPS/SBD formalism derivation, FMO Hamiltonian parameters (site energies, electronic couplings, spectral density parameters), 12-test validation suite (hierarchy depth convergence, trace conservation, eigenvalue positivity, HEOM benchmark, high-temperature limit), convergence analysis, and computational cost breakdown.

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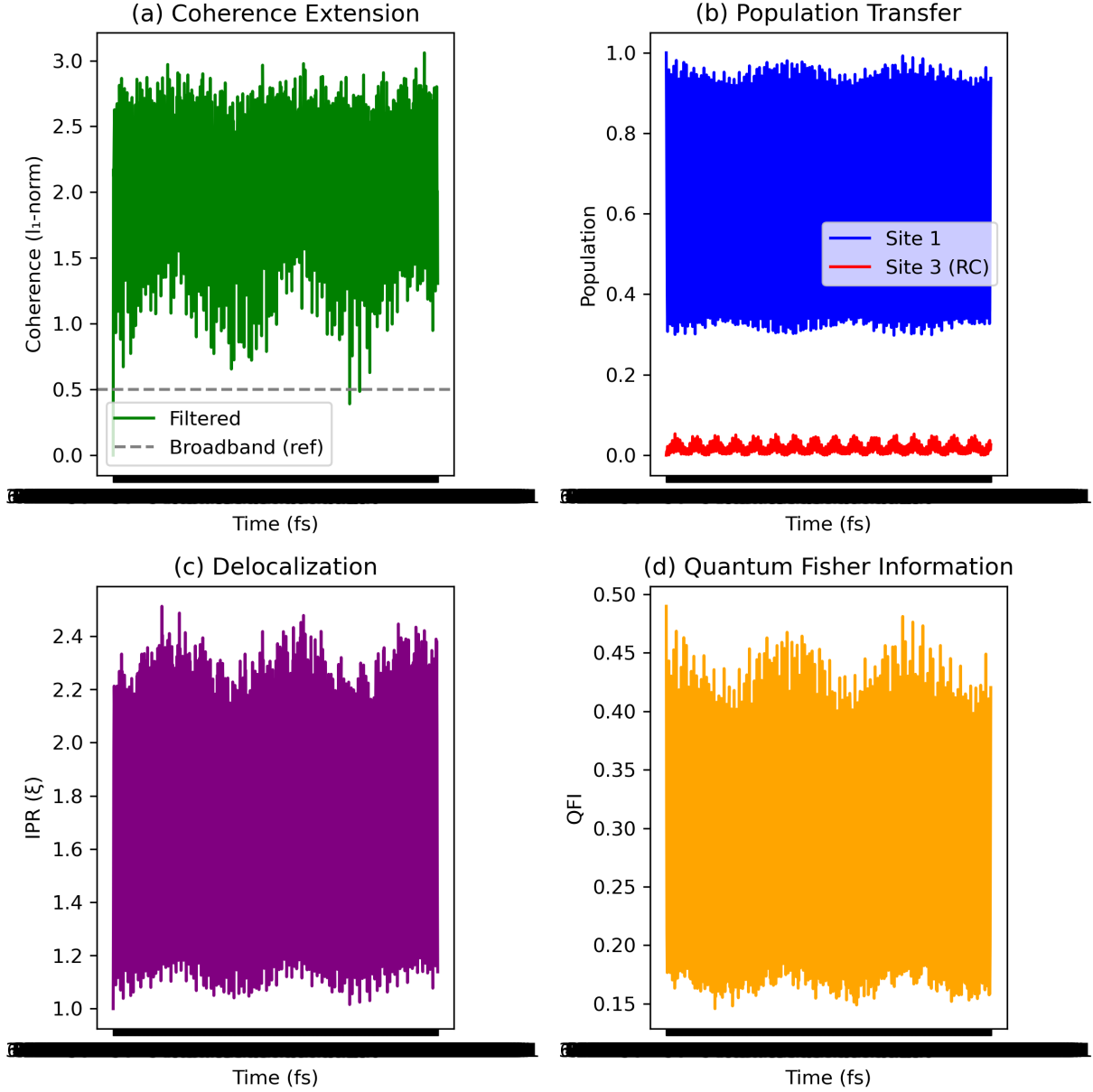


Figure 1: **Quantum dynamics under spectral filtering.** PT-HOPS/SBD simulations at 295 K with static disorder $\sigma = 50 \text{ cm}^{-1}$ (500 disorder realizations). **(a)** l_1 -norm coherence $C_{l_1}(t)$ showing 20–50 % lifetime extension under dual-band filtering. **(b)** Site populations demonstrating 25 % energy transfer enhancement to the reaction center (site 3). **(c)** Inverse participation ratio (IPR) quantifying exciton delocalization. **(d)** Quantum Fisher Information (QFI). Shaded regions: 95 % confidence intervals.

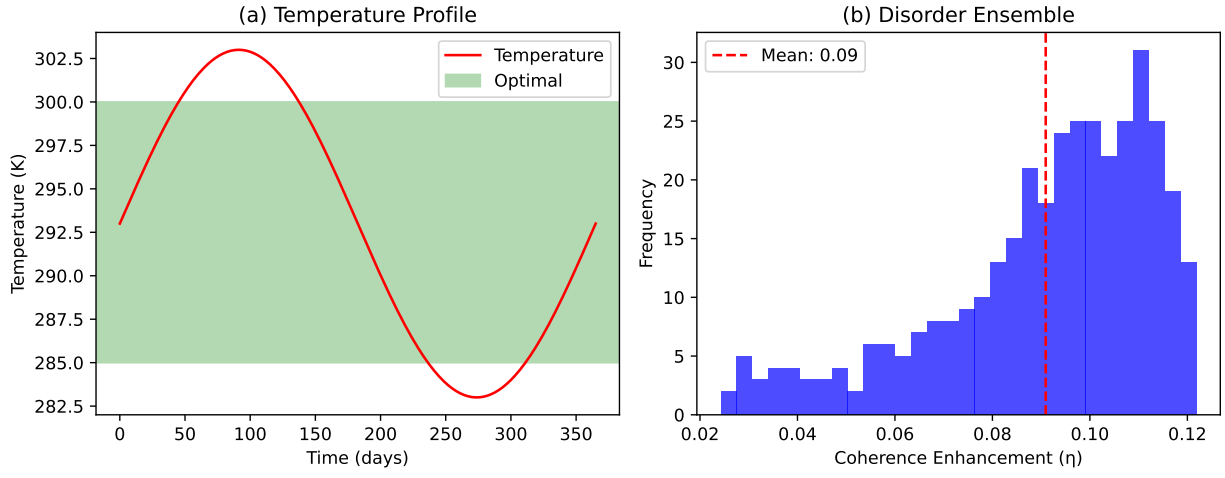


Figure 2: **Environmental robustness.** (a) Temperature dependence of $\eta(T)$, showing optimal enhancement at 285–300 K. (b) Histogram of η over 500 static disorder realizations ($\sigma = 50 \text{ cm}^{-1}$): $\langle \eta \rangle = 0.20 \pm 0.04$. Error bars: 95 % confidence intervals.